Course of Mechatronics - Exercitation 1: DC Motor Simulation

A DC Motor can be seen as a set of electrical components and mechanical parts. The angular velocity depends on the applied voltage; the generated torque depends on the current.

About the **electrical entities**:

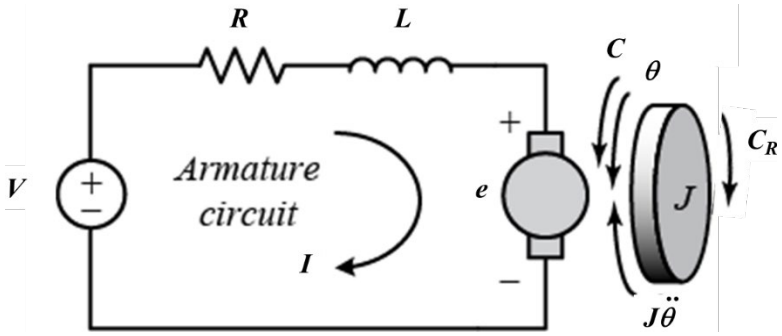
V: is the voltage source applied to the armature circuit of the motor *[V]*;

R: is the resistor of the armature of the motor *[Ω]*;

L: is the inductor of the armature of the motor *[H]*;

e: is the back electromotive force *[V]*;

I: is the current of the armature *[A]*;



About the **mechanical entities**:

C: is the generated torque [Nm];

C_R: is the resistant torque [Nm];

J: is the momentum of inertia of the rotating masses [kgm^2];

For this system, four characteristic relations can be written:

1. the armature circuit equation
2. the equilibrium equation, as regards the rotation;
3. $e(t) = K_e \cdot \omega$
4. $C = K_t \cdot I(t)$

Where: **K_e**: is the back electromotive force constant [*Vs/rad*];

K_t: is the torque constant [*Nm/A*].

Given the following values:

V=0÷12 V	**R=4.4 Ω**	**L=6.2 mH**	**J=0.0000016 kgm^2**	**K_e=0.025 Vs/rad**	**K_t=2.5 Ncm/A**

and in 3 different conditions of the system:

C_R_stall = 6.6 Ncm	**C_R_rated = 2.0 Ncm**	**C_R_noLoad = 0.0 Ncm**

the **objectives of the exercitation** are:

1. Write the two remaining characteristic relations: the armature circuit equation and the equilibrium equation;
2. Then, find the behaviour of I(t) and ω(t) for each of the 3 C_R conditions, in the range of V, according to the:
   i. Direct Solution of the differential equation method;
   ii. States Space method;
   iii. Transfer Function method.

   implemented in Simulink;
3. Graphically, compare the results obtained by the previous methods;
4. Finally, find the characteristic curve of the motor, ω=ω(V, C_R), in the following terms:
   i. ω=ω(C_R) for V = [0 2 4 6 8 10 12] V;
   ii. ω=ω(V) for C_R = [0 1 2 3 4 5 6] Ncm.

Put all the contents of the present exercitation in a technical document to be delivered and discussed with the Lecturer at the end of the course. Students must prepare the technical paper as a result of the workgroup.